

Chapter 1 — Exploring a dataset

Discover whether a concrete file meets the quality bar

Learning objectives

- State three early questions exploration should answer
- Match common tools to dataset size and structure
- Outline a first-pass profiling pattern (summaries + visuals + missingness)

Why explore first?

- Turns an unfamiliar file into a mental model
- Surfaces seasonal patterns, outliers, missing fields
- Shapes which quality issues to prioritize
- Does not replace formal cleaning—reveals where it is needed

Three early questions

- What is in the table?
- What looks suspicious?
- What analysis goals remain feasible given defects?

Tools for exploration

- Spreadsheets — small tables, quick looks
- SQL — filter and aggregate relational stores
- pandas — reproducible profiling scripts
- Tableau / Power BI — shared interactive views

Example 1.20 — Household incomes EDA

```
$ cd modules/chapter1/example20/ && bash run.sh
```

Takeaways

- Explore before modeling to set expectations
- Tool choice follows size, structure, and audience
- Document what you inspected and what you will fix